LAB 02 SOLUTIONS:

Exercise 01:

Make a program that takes inputs like height, width and depth from user and then calculate volume of the cube: volume = height ∗ width ∗ depth After calculating volume of cube, compare it with following ranges and print the relevant label ...

CODE:

```
# LAB EX 01: (i)

height = float(input("Enter height in cm : "))
width=float(input("Enter width in cm : "))
depth=float(input("Enter depth in cm : "))
volume= height*width *depth

print("\nVolume of object: ",volume)

if(volume <1):
    print("Label : Invalid inputs.")
elif (volume <= 10):
    print("Label : extra small")
elif (volume <= 25):
    print("Label : small")
elif (volume <= 75):
    print("Label : Medium")
elif (volume <= 100):
    print("Label : Large")
elif (volume <= 250):
    print("Label : Extra Large")
else :
    print("Label : Extra Extra Large")
```

OUTPUT:

```
Enter height in cm : 7
Enter width in cm : 3
Enter depth in cm : 5

Volume of object:  105.0
Label : Extra Large
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

EXERCISE 02:

In a company ,worker efficiency is determined on the basis of the time required for a worker to complete a particular job.If the time taken by the worker is between 2-3 hours then the worker is said to be highly efficient. If the time required by the worker is between 3-4hours,then the worker is ordered to improve speed. If the time taken is between 4-5

hours ,the worker is given training to improve his speed ,and if the time taken by the worker is more than 5 hours ,then the worker haas to leave the company, If the time taken by the worker is input through the keyboard,find the efficiency of the worker.

CODE:

```
26  # ii
27  worker_time = float(input("Enter time taken by worker : (in hours)  "))
28  if worker_time <0:
29      print("Invalid input")
30  elif( worker_time <= 3):
31      print("Highly efficient")
32  elif( worker_time <= 4):
33      print("Improve your speed")
34  elif( worker_time <= 5):
35      print("Take the training to improve speed")
36  else:
37      print("You may leave the company ! ")
38
```

OUTPUT:

```
Enter time taken by worker : (in hours)  3
Highly efficient
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

EXERCISE 03:

)The program must prompt the user for a username and password. The program should compare the password given by the user to a known password. If the password matches, the program should display “Welcome!” If it doesn’t match, the program should display “I don’t know you. Note: the password should not be case sensitive

CODE:

```
39  # iii
40  username= input("Enter your username : ")
41  password = input("what is the password ? ")
42  real_password= "abc$123"
43
44  if(password.lower() == real_password): #as in question->password not case sensitive
45      print("Welcome ! ")
46  else:
47      print("I don't know you. ")
48
```

OUTPUT: (of both cases)

```
Enter your username : samia qadri
what is the password ? abc
I don't know you.                                          python -u "c:\Users
I\uni_labs\lab02\tempCodeRunnerFile.py"AI\uni_labs\lab02>
Enter your username : samia qadri
what is the password ? AbC$123
Welcome !
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

EXERCISES:

1. Create a loop that counts from 0 to 100

```
for n in range(101):
    print(n)
    #n+=1
```

OUTPUT: (0 10 100 counting)

```
0
1
2
3
4
5
6
7
```

2. Make a multiplication table using a loop

```
68   # 2. Make a multiplication table using a loop
69   num=int(input("Enter number whose table you want : "))
70   # n=1
71   for n in range(1,11):
72       print(num ," * ",n," = " ,num*n)
```

Output:

```
I\uni_labs\lab02\tempCodeRunnerFile.py"
Enter number whose table you want : 3
3  *  1  =  3
3  *  2  =  6
3  *  3  =  9
3  *  4  =  12
3  *  5  =  15
3  *  6  =  18
3  *  7  =  21
3  *  8  =  24
3  *  9  =  27
3  *  10  =  30
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

3. Output the numbers 1 to 10 backwards using a loop

```
n=10
while(n>0):
    print(n)
    n-=1
```

OUTPUT:

```
10
9
8
7
6
5
4
3
2
1
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

4. Create a loop that counts all even numbers to 10

```
count=0
print("even numbers are as: ")
for i in range(0,11):
    if(i% 2==0):
        count+=1
        print(i)
print("total even numbers are : ",count)
```

OUTPUT:

```
even numbers are as:
0
2
4
6
8
10
total even numbers are :  6
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

5. Create a loop that sums the numbers from 100 to 200

```
sum=0
for i in range(100,201):
    sum +=i
print("Sum of numbers between 100 to 200 is : ",sum)
```

OUTPUT:

```
Sum of numbers between 100 to 200 is :  15150
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

(iii) Try the exercise below:

1. Make a program that lists the countries in the set below using a while loop.

clist = ["Canada","USA","Mexico"]

```
clist = ["Canada","USA","Mexico"]
i = 0
while i< len(clist):
    print(clist[i])
    i +=1
```

OUTPUT:

```
Canada
USA
Mexico
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab02>
```

2. What's the difference between a while loop and a for loop?

3. Can you sum numbers in a while loop?

4. Can a for loop be used inside a while loop?